DS - Assignment - 2 24/3/2022

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1. What is data warehouse? Write its key features

DW provides architectures and tools for business executives to systematically organize, understand and use their data to make strategic decision.

Key features are:

i) Subject-oriented - A DW is organised around major subjects such as customer, supplier, product and sales. Rather than concentrating on the day-to-day operations and transaction processing of an organisation, a data ~~waref~~ warehouse focuses on modeling and analysis of data for decision makers.

ii) Integrated: A DW is usually constructed by integrating multiple heterogeneous sources, such as relational db, flat files and online transactions records.

iii) Time variant: data are stored to provide information from a historical perspective.

iv) Non-volatile: A DW is always a physically separate store of data transformed from the application data found in operational environment.

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2) Explain difference b/w Operational DB and DW

Operational database	Data warehouse
1) Customer-oriented	1) Market-oriented
2) Used for transaction & query processing.	2) used for data analysis by knowledge workers.
3) Manages current data	3) Manages large amount of historical data.
4) Adopts an ER data model and an application-oriented database design.	4) Adopts either a star or snowflake model and a subject-oriented db design
5) Consists mainly of short, atomic transactions.	5) Consists mainly read-only operations.
6) Size of db - 100MB to GB	6) Size - 100GB to TB

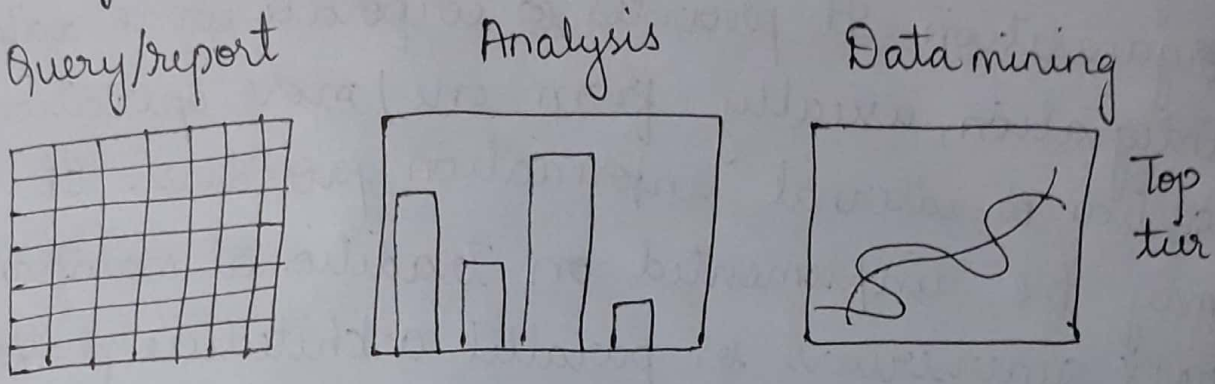
3) Discuss 3-tier dW architecture

i) Bottom tier: is a warehouse db server that is always a relational db system. Data from operational db and external sources are extracted using application program interface known as gateways. Gateways is supported by underlying DBMS and allows client program to generate SQL code to be executed at a server.

ii) The middle tier: is an OLAP server that is typically ~~build~~ implemented using either a relational OLAP(ROLAP) or multidimensional OLAP)

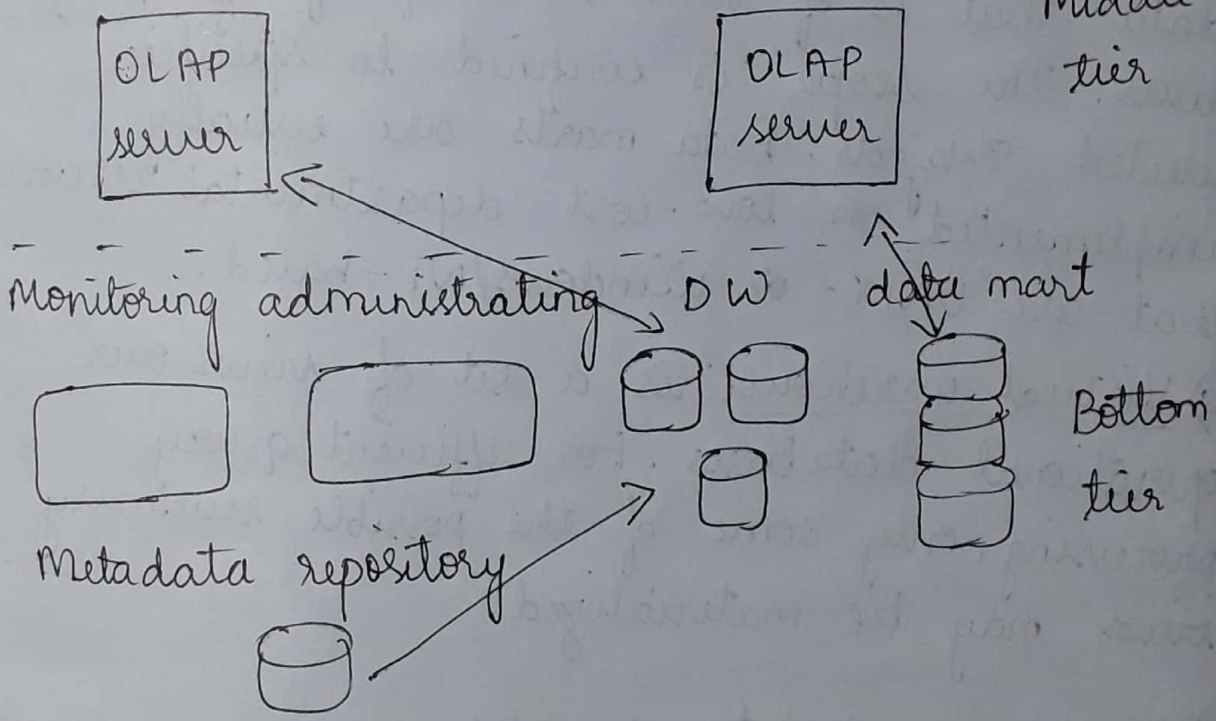
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3) Top tier : is a client, which contains query and reporting tools, analysis tools and/or data mining tools.



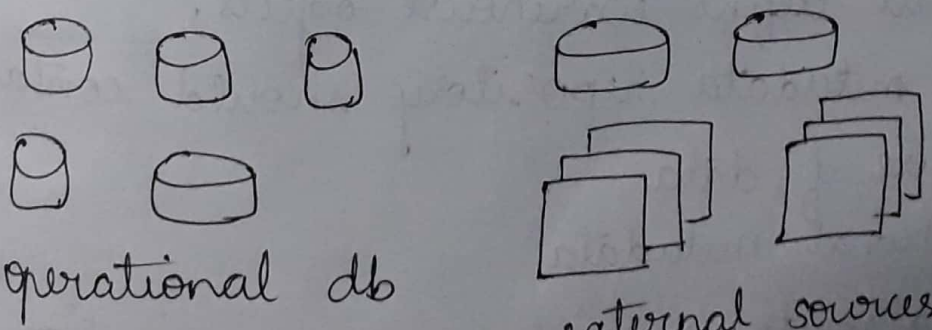
↑ output

Middle tier



↑ extract, clean, transform, load, refresh

Data



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4) Explain the following models:

a) Enterprise warehouse: collects all of the information about subjects spanning the entire organisation. It provides corporate-wide data integration, usually from one / more operational system or external information providers. It may be implemented on traditional mainframes UNIX superservers or parallel architecture platforms

b) Data mart: contains a subset of corporate-wide data that is of value to a specific group of users. The scope is confined to specific selected subjects. Data marts are usually implemented on low-cost departmental servers that are UNIX- or Windows/NT-based.

c) Virtual warehouse: is a set of views over operational databases. For efficient query processing, only some of the possible summary views may be materialized.

6) Explain metadata repository.

Metadata are data about data. When used in data warehouse, metadata are the data that define warehouse objects.

A metadata repository should contain

-> structure of data
-> Operational metadata
-> The algorithms used for summarization

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→ The mapping from the operational environment to data warehouse
→ data related to system performance
→ Business metadata.

11) Explain OLAP operations

a) Roll-up: This operation is also called as drill-up operation by some vendors. The roll-up operation performs aggregation on a data cube, either by climbing up a concept hierarchy for a dimension or by dimension reduction. When roll-up is performed by dimension reduction, one or more dimensions are removed from given cube.

b) drill-down: is a reverse of roll-up. It navigates from less detailed data to more detailed data. Drill-down can be realized by either stepping down a concept hierarchy for dimension or introducing additional dimension.

c) Slice and dice: The slice operation performs a selection on one dimension of given cube, resulting in a subcube. The dice operation defines subcube by performing a selection on two or more dimension.

d) Pivot: also called as rotate. Pivot is a visualization operation that rotates the data axes in view to provide an alternative data presentation.

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Additional drilling operations:

drill across: executes queries involving more than one fact table.

drill-through: operations uses relational SQL facilities to drill through the bottom level of a data cube down to its back-end relational tables.

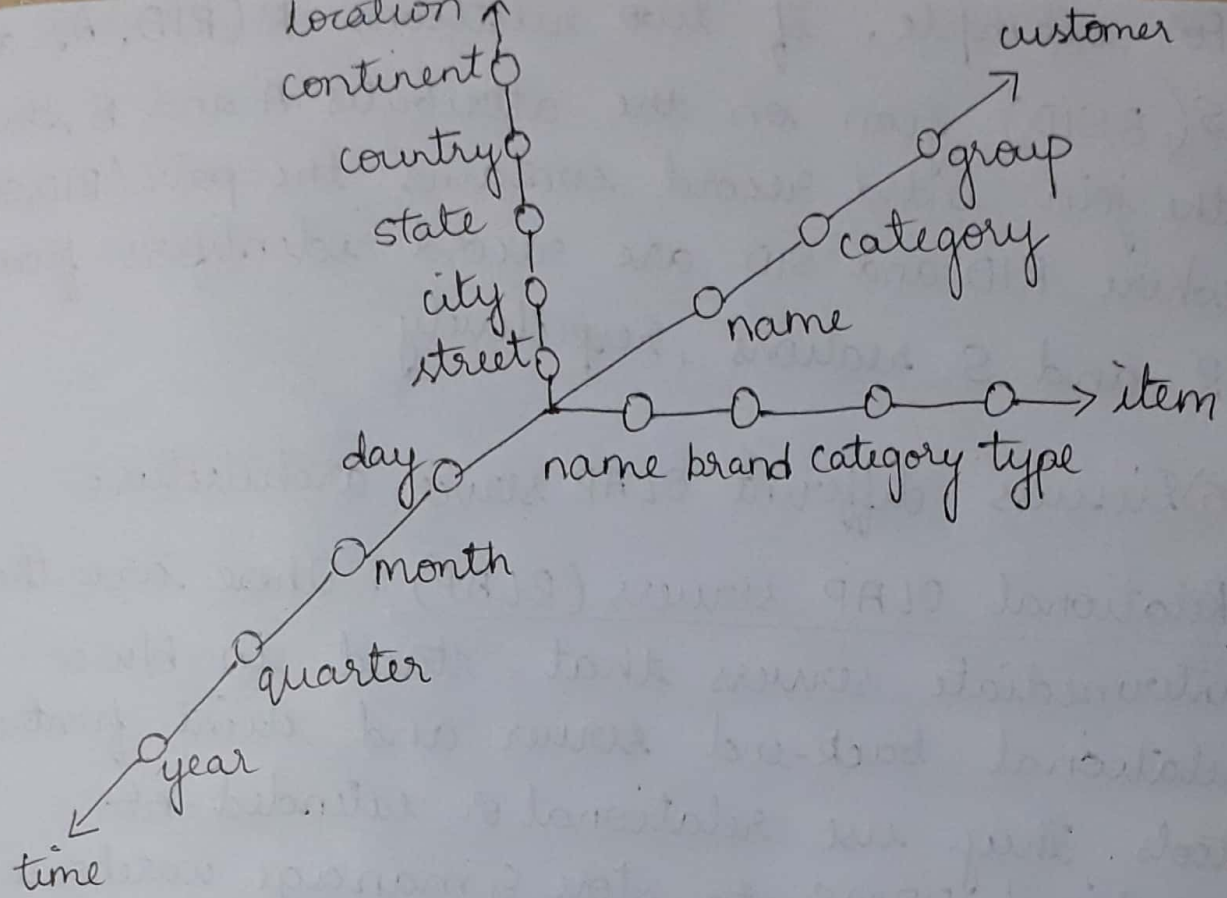
8) Discuss snowflake schema, fact constellation, starnet query model of multidimensional data model.

Snowflake schema: is a variant of the star schema model, where some dimension tables are normalized, thereby further splitting the data into additional tables. The resulting schema graph forms a shape similar to a snowflake.

Fact constellation: Sophisticated application may require multiple fact tables to share dimension tables. This kind of schema can be viewed as a collection of stars known as a galaxy schema or a fact constellation.

Starnet query model: The querying of multi-dimensional db can be based on starnet model. A starnet model consists of radial lines emanating from a central point, where each line represents a concept hierarchy for a dimension. Each abstraction level in the hierarchy is called footprint. These represent the granularities available for use by OLAP operations.

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14) Discuss indexing in OLAP data for data accessing DW

To facilitate efficient accessing, most DW system index structures and materialized views using cuboids. Indexing OLAP data by Bitmap indexing is popular in OLAP products because it allows quick searching in data cubes. The bitmap index is an alternative representation of the record-ID (RID) list.

Join indexing: gained popularity from its use relational db query processing. Traditional indexing maps the value in a given column to a list of rows having that value. In contrast, join indexing registers the joinable rows of 2 rations from a relational db.

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For example, if two relations R(RID, A) and S(B, SID) join on the attributes A and B, then the join index record contains the pair (RID, SID) where RID and SID are record identifiers from R and S relations, respectively.

15) Discuss different OLAP server architecture

Relational OLAP servers (RLAP): These are the intermediate servers that stand in b/w a relational back-end server and client front-end tools. They use relational or extended-relational DBMS to store & manage warehouse data and OLAP middleware to support missing pieces.

Multidimensional OLAP server (MOLAP): These servers support multidimensional views of data through array-based multidimensional storage engines. They map multidimensional views directly to data cube array structures.

Hybrid OLAP server (HOLAP): approach combines ROLAP and MOLAP technology, benefiting from greater scalability of ROLAP and faster computation of MOLAP.

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